Private federated brain tumour MRI classification, audited

Clean benchmark

Figshare, SARTAJ, Br35H

7828 labelled images

↓

3845 after hashing
patient-disjoint splits

66 of 813 test images
are mirrored copies

↓

747 clean test images

Private training

DP-SGD on every client
image or patient as unit

record level, $\varepsilon = 8$
balanced-accuracy cost
0.076 (FedAvg),
0.130 (central)

patient level needs
$\varepsilon = 64$ to exceed it

Leakage audit

LiRA, 16 shadow models

TPR at 1% FPR
no DP 0.108
DP 0.010 to 0.017 (chance)

unclipped weight noise
patient AUC 0.745

Explanation audit

Grad-CAM vs expert masks

cross-seed consistency
0.700 (no DP)
to 0.235 ($\varepsilon = 1$)

tracks accuracy, $r = 0.798$
noise adds nothing
beyond accuracy